CLASSIFICATION

TRAIN/VALIDATION/TEST AND FEATURES SELECTION

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Train/Validation/Test and Features Selection

The following concepts will be introduced:

- ✓ **Regularization**
- ✓ **Train/Validation/Test**
- ✓ **Features Selection**

REGULARIZATION

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Some Machine Learning models allow hyper-parameters, e.g. regularization in Neural Networks:

$$E(\mathbf{w},\lambda) = \frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2 + \frac{\lambda}{2}\sum_{j=1}^{K} w_i^2$$

K; number of free parameters (weights + thresholds) of the neural network

λ; regularization parameter.

How to select a proper value for the regularization parameter λ ?

REGULARIZATION

Use the Test dataset to “optimize” the regularization parameter λ ?

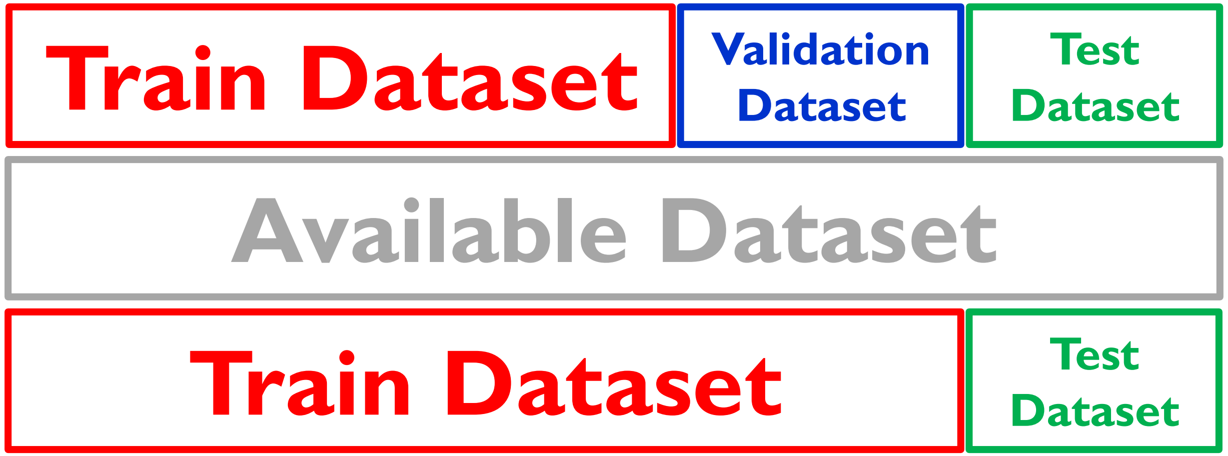
Available Dataset

Train Dataset | Test Dataset

You are selecting the value of the regularization parameter λ to optimize some performance measure of the neural network.

You get a biased estimate of the optimized performance measure, i.e. you are typically over optimistic about the performance that the neural network achieve on the test dataset, i.e. you could be overfitting!!!

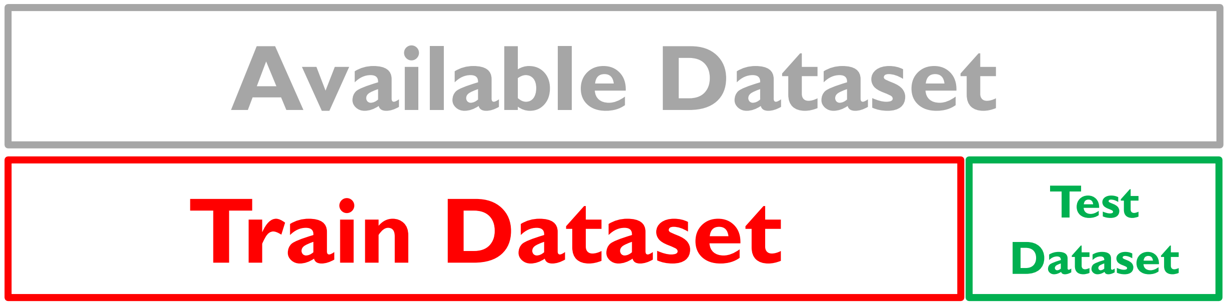
When the Available Dataset is “small” (whatever it means), it may not be possible or undesirable to apply it!!!!



- The regularization parameter λ is optimized using the Validation Dataset.
- The Test Dataset is used to provide an unbiased estimation of the performance measure.
- This can be done by Hold-out, Iterated Hold-out, and Cross Validation.

Feature Selection – Filter

When using Filters (uni-variate and multi-variate), Features Selection is typically performed using the Train/Test scheme.

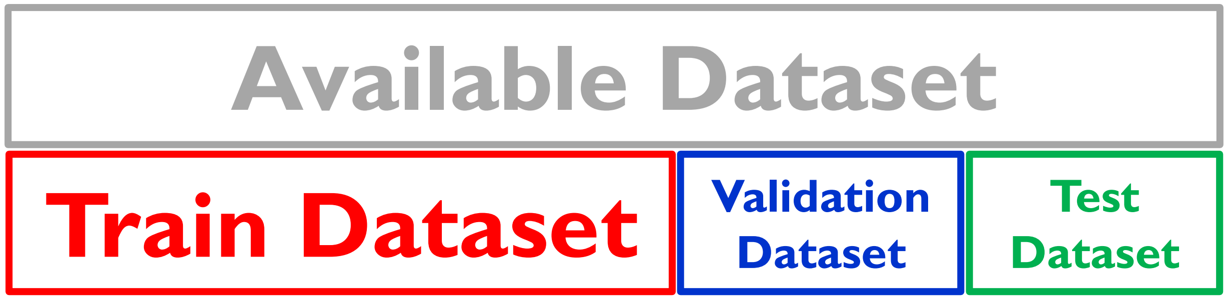


- Relevance and/or redundancy are estimated using the Train Dataset.
- Once the attributes have been selected, the Train Dataset is used to learn a classifier.
- The performance estimate is obtained by applying the classifier to the Test Dataset.

FEATURE SELECTION – WRAPPER

When using Wrappers, Features Selection is typically performed using the Train/Validation/Test Scheme.

- Relevance and/or redundancy are estimated using the Train Dataset and the Validation Dataset.



- The Validation Dataset is used optimize the performance measure when different attributes are used by the classification model, i.e. the **Wrapper**!!!
- Selecting the optimal subset of attributes is the same as selecting the optimal value of the regularization parameter λ.
- Once the attributes have been selected, the Train Dataset and the Validation Dataset are merged together to learn the classifier (typically (not mandatory) the same type of classifier as the one used for attribute selection, i.e. the **Wrapper**!!!).
- The performance estimate is obtained by applying the classifier to the Test Dataset.